

# Tobacco Smoking

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 Adapted from: [Tobacco Smoking](#)

Listen to this Brick

0:00 / 34:44 1x

Learning Objectives (5)

Versions

*After completing this brick, you will be able to:*

- 1 Describe tobacco smoking and vaping and their epidemiology.
- 2 Identify tobacco-related adverse health outcomes.
- 3 Explain the pathophysiology of tobacco smoking-related disease.
- 4 Describe how to screen patients for tobacco use and exposure.
- 5 Outline management options for smoking cessation including pharmacological interventions and secondary prevention of smoking-related illness.



CASE CONNECTION

JB, a 24-year-old artist, has been vaping tobacco since age 17. He grew up in a house where both his parents smoked cigarettes and was exposed to passive smoke on a regular basis. His father was recently diagnosed with chronic obstructive pulmonary disease (COPD). During his annual check-up, JB tells his primary care physician that he is thinking about quitting smoking because he fears the long-term health consequences. But this is difficult because his close friends vape as a social activity. He asks for more information on the risks of smoking and how he can stop.

What advice should the doctor give JB? Consider your answer as you read, and we'll revisit JB at the end of the brick.

[GO TO CONCLUSION](#) ↓

## What Are Smoking and Vaping? (LO#1)

Smoking and vaping are associated with serious health risks to multiple body systems, particularly lung and cardiovascular disease, as well as multiple forms of cancer. People who smoke tobacco have a three-fold heightened mortality rate compared to those who have never smoked. As a result, it's critical for all health care professionals to discourage smoking and counsel and educate their patients on its many harmful effects and effective ways to quit. Smoking's social impact and high illness burden affect legislation, environmental policy, and health policy.

Smoking tobacco dates back to Mayan and Aztec times, and it was introduced to Europe in the sixteenth century by Spanish conquerors. It is now a worldwide practice, largely due to the addictive effects of **nicotine**, a chemical compound inhaled in tobacco smoke. Unfortunately, cigarette tars and chemicals contain numerous carcinogenic compounds. This brick will look more closely at smoking and its effects on human health.

## Smoking vs Vaping

Smoking refers to inhaling the fumes from tobacco combustion (burning) within a cigarette. The addictive effect comes from **nicotine** inhaled during the process. The inhaled tars, carbon monoxide, and oxidizing gasses cause most of its long-term health consequences.

Nicotine also can be ingested orally (eg, chewing tobacco and nicotine gums used for smoking cessation) or inhaled by **vaping** (sometimes referred to as e-cigarettes), which involves heating, atomizing, and inhaling a nicotine-containing liquid. Vaping produces particulates, not a true vapor, so is not well named (Figure 1).



Figure 1 - Vape

Because vaping smoke does not contain most of traditional tobacco smoking's byproducts, it has become increasingly popular in recent years, especially among young people who perceive it to be risk-free. But vaping is not harmless.

While long-term survival statistics are not yet available, vaping can cause nicotine poisoning (especially in children). In addition, there have been numerous reports of death as a result of acute lung injury (termed vaping-associated pulmonary injury [VAPI]). This type of lung injury includes eosinophilic pneumonia as well as lipid pneumonia with lipid laden macrophages, with rapid progression to acute respiratory syndrome (ARDS) and death. These deaths primarily occur in young people and in those who also use tetrahydrocannabinol (THC) electronic products. Inhaling propylene glycol, vitamin E acetate, and other chemicals in the vaping fluid may also have adverse effects.

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cigarette that evaporates nicotine-containing liquid.

cigarette while vaping involves inhaling nicotine vapor from an e-  
Tobacco smoking entails inhaling nicotine smoke from a burning

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## Epidemiology

Approximately 30 million people in the United States smoke (9%-10% of the population), and about half of them have a smoking-related disease. This number has declined from 20% since 2005, largely due to antismoking campaigns.

Worldwide it is estimated that about 20% of the population smokes. Smoking rates are highest in males, young adults, those with lower education status, those with a partner or other household member who smokes, and coexisting mental health disorders, eg, depression.

Vaping is common in the young, with a 2022 study showing that 14% of US high school students (ages 15-18) and 3% of middle school students (ages 12-15) vape, totaling more than 2.5 million young users.

## What Are Common Health Effects of Smoking? (LO#2)

Adverse effects of smoking can be acute or chronic and are related to the many chemicals in cigarette smoke, as well as to nicotine's stimulant and addictive effects. Tobacco use has the potential to harm multiple organ systems (Figure 2).

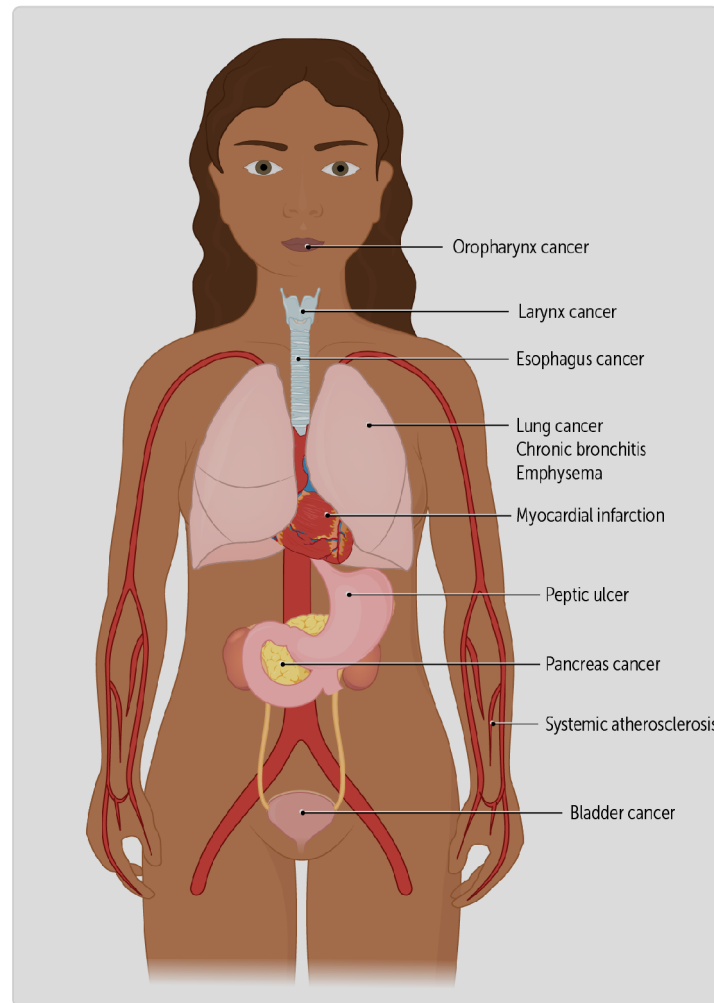


Figure 2 Common adverse effects of tobacco smoking

## Acute Adverse Effects

Many substances in inhaled cigarette smoke are lung irritants, so smoking can exacerbate pre-existing lung diseases like asthma or chronic obstructive pulmonary disease (COPD). This causes increased coughing and wheezing.

Nicotine's acute effects include sympathetic stimulation, leading to tachycardia, increased blood pressure, and vasoconstriction. These effects can aggravate symptoms of ischemic cardiac disease (causing chest pain) or peripheral arterial disease (causing leg pain). Nicotine poisoning in children (increasingly common with vaping) can cause vomiting, nausea, tachycardia or, rarely, seizures, coma, and cardiac arrest.

## Chronic Adverse Effects

Chronic tobacco use results in pulmonary, cardiovascular, and oncologic (cancer-related) diseases.

**Pulmonary.** Chronic pulmonary disease is strongly connected to smoking, particularly **COPD**, which is characterized by progressive cough, wheezing, dyspnea, and hypoxia. COPD encompasses emphysema (breakdown of alveoli and lung destruction) and bronchitis (airway inflammation with a chronic productive cough).

Smoking is associated with most forms of lung cancer, causing 70% of cases. Those who smoke have a 10- to 30-fold higher lung cancer risk than nonsmokers.

In the United States, smoking cessation has reduced lung cancer mortality by nearly 80% over the past 50 years, making it one of the most successful medical advances in modern history.

Other smoking-related pulmonary diseases include infections (pneumonia, tuberculosis, influenza) and idiopathic pulmonary fibrosis. It also is linked to several uncommon interstitial lung diseases, including desquamative



interstitial pneumonia (DIP), respiratory bronchiolitis-interstitial lung disease, and pulmonary Langerhans cell histiocytosis.

**Cardiovascular.** Smoking is a major and modifiable risk factor for **atherosclerotic cardiovascular disease**, including both coronary artery disease (myocardial infarction, angina), heart failure, stroke, and peripheral artery disease. Passive (second-hand) smoking is also correlated with increased risk of cardiovascular disease. Smoking cessation significantly reduces the risk of coronary atherosclerosis and myocardial infarction.

Smoking also increases the risk of **thromboembolic disease** such as deep venous thrombosis (DVT) and pulmonary embolus (PE). Thromboangitis obliterans (Buerger disease) is a small- and medium-vessel inflammation that is seen largely in people who smoke tobacco. It causes thrombosis of small blood vessels, accompanied by acute pain in the fingers and toes. This disease primarily affects young men under the age of 40 years who are heavy smokers, but its mechanism is unknown.

**Nonpulmonary Cancer.** Those who smoke tobacco have a 20-fold increased risk of developing cancer when compared to those who are nonsmokers. The products within tobacco smoke induce gene mutations that cause unregulated cell growth (neoplasia). Besides lung cancer, smoking has been linked to a variety of other cancers:

- Gastrointestinal (GI) tract: oropharynx, esophagus, stomach, pancreas, colon, liver
- Genitourinary (GU) tract: kidney (renal cell carcinoma); bladder (urothelial cell carcinoma)
- Larynx
- Lip
- Leukemias and lymphomas

The nonsmoking-connected exceptions include cancers of the brain and reproductive tract.

Stopping smoking after diagnosis increases survival rates in many of these cases.

**Gastrointestinal Disease.** Gum disease (periodontal disease), peptic ulcers, and pancreatitis.

**Renal.** Increased risk of developing chronic kidney disease (CKD) and accelerating the progression of preexisting CKD.

**Reproductive.** Increased pregnancy complications and erectile dysfunction. It increases the risk of male and female infertility.

**Skeletal.** Increased bone demineralization, leading to increased risk of osteoporosis and hip fractures.

**Ocular.** Cataracts, glaucoma, and age-related macular degeneration.

**Endocrine.** Increased risk of developing type 2 diabetes mellitus, likely due to decreasing insulin sensitivity.

**Psychiatric.** Patients with mental health disorders smoke more and often develop nicotine addiction. Smoking may aggravate some disorders, particularly psychotic disorders (eg, schizophrenia). Nicotine alters the

release of many neurotransmitters, and there may be common genetic links between schizophrenia and smoking.

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kidney, leukemia, and lymphomas.

include cancers of the lung, gastrointestinal tract, larynx, lip, bladder,

Examples of cancers strongly associated with tobacco smoking

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## What Is the Pathophysiology of Tobacco Smoking-Related Disorders? (LO#3)

We can differentiate the adverse effects of tobacco smoke into those caused by nicotine and those caused by other inhaled toxins.

### Effects of Nicotine

Nicotine is a natural alkaloid derived from the nightshade plant family that has behavioral and physiological effects on humans. Its strong behavioral effects are generally experienced as pleasurable by most people. This is a big reason why smoking cessation is often extremely hard for long-time smokers.

Nicotine is classified as a highly **addictive** substance. Addiction is characterized by a chronic, compulsive need for increasingly more of the substance. Chronic use also causes physiological **dependence**, with withdrawal symptoms such craving, insomnia, anxiety, depression, and hunger. Chronic nicotine ingestion also induces **tolerance**, since increasing smoking is needed to achieve the pleasurable effects and prevent withdrawal symptoms.

Nicotine has several physiologic effects, mostly related to its binding to the nicotinic acetylcholine receptors (nAChRs). These receptors are normally activated by binding to the neurotransmitter acetylcholine, but nicotine stimulates them as well. Consequently, nicotine is a **sympathomimetic** substance, stimulating the sympathetic nervous system, leading to tachycardia and increased blood pressure. Nicotine suppresses appetite, leading some patients to continue smoking to prevent weight gain. Nicotine also induces endothelial cell injury, tissue hypoxemia, and increased platelet activity.

Nicotine exerts multiple effects in the brain (Figure 3).

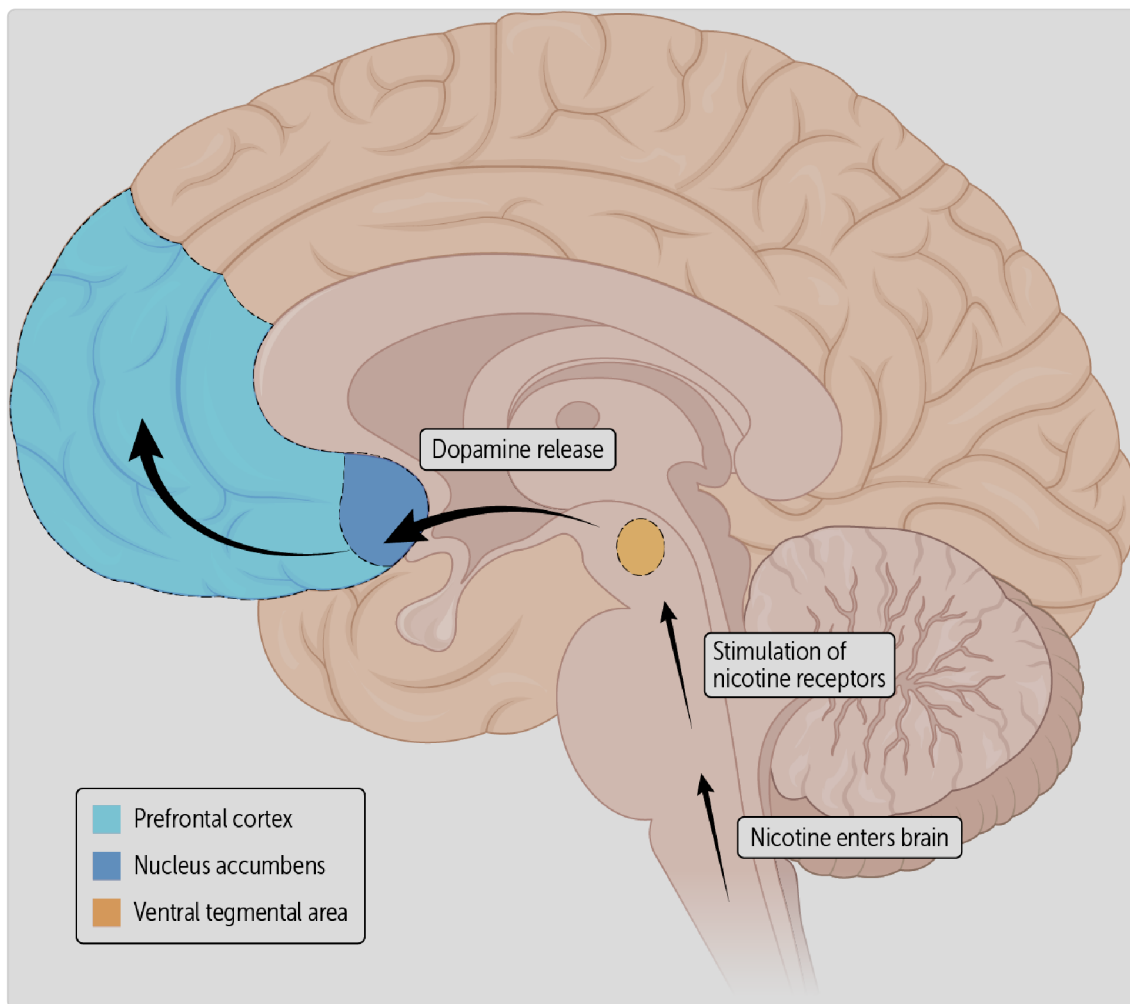


Figure 3 Nicotine stimulation of the brain reward pathway

These effects include:

- Direct stimulation of the brain's reward center, which is part of the basal ganglia in the forebrain and midbrain reaching from the ventral tegmental area of the midbrain to the nucleus accumbens. There, dopamine release stimulates the pleasure and impulse control centers in the prefrontal cortex, which are responsible for nicotine's addictive effects.

- Stimulation of serotonergic release throughout the brain, implicated in addiction.
- Inhibition of the enzyme monoamine oxidase, which raises levels of brain neurotransmitters like norepinephrine, serotonin, and dopamine.
- Elevated brain levels of endorphins and corticosteroids. Stimulation of the hypothalamic-pituitary pathway increases pituitary adrenocorticotrophic hormone (ACTH) release, leading to adrenal gland cortisol secretion. The increased cortisol boosts alertness in users.

used.

Serotonin, dopamine, and endorphins are elevated when nicotine is

## Effects of Other Tobacco Smoke Toxins

Aside from nicotine, tobacco smoke contains more than 7000 different chemicals, of which at least 250 have been confirmed to be harmful. These include hydrocarbons, radioactive components, and metals. More than 60 are known to be carcinogenic. Some of these, and where to find them (apart from cigarette smoke), are shown in [Figure 4](#).

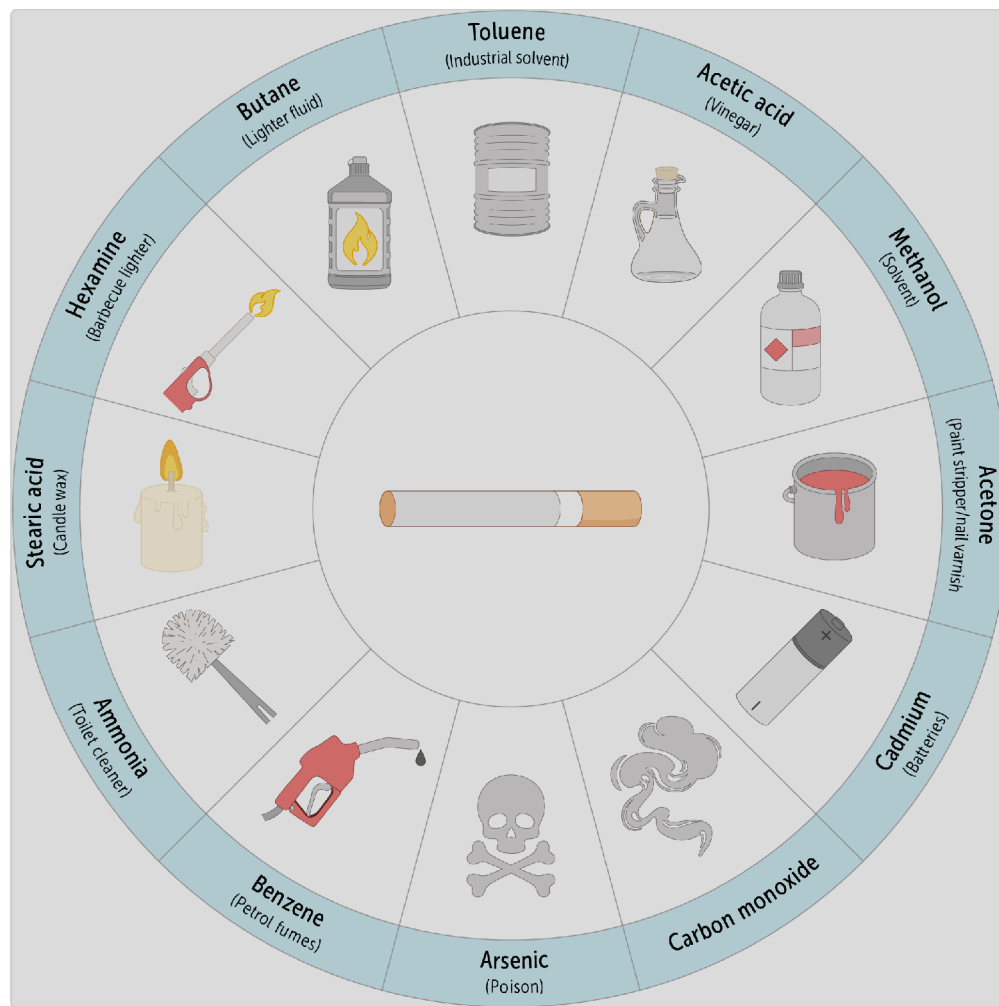


Figure 4 Toxins in tobacco smoke

How do these toxins cause smoking-related disease? The effects are complex, but a few that are understood include:

**Atherosclerosis.** Smoking enhances the formation of atherosclerotic plaques in blood vessels, the main mechanism for the development of coronary and peripheral arterial disease. Smoking directly induces **endothelial cell damage**, necrosis, and apoptosis, in part by generation of reactive oxygen species.

**Hypercoagulability.** Smoking increases platelet aggregation and damages blood vessel walls, increasing the likelihood of clots forming in blood vessels.

This decreases blood flow to the tissues, amplifying the effect of atherosclerosis.

**Impaired Red Blood Cell (RBC) Oxygen Release to Tissues.** The carbon monoxide in cigarette smoke impairs both oxygen binding to RBC and the release of the bound oxygen to tissues. This can be a problem in patients who have ongoing atherosclerosis caused by smoking, which reduces blood flow to tissues.

**Inflammation and Immune Abnormalities.** In smokers, **inflammation increases** throughout the body, measured by increased levels of c-reactive protein (CRP). The multiple mechanisms include activation of oxidation-sensitive cellular signaling pathways, increased cytokine release (eg, IL-8, TNF-alpha), activation of inflammatory genes, peroxidation of lipids, and recruitment of inflammatory cells from the microcirculation of the lung.

**Innate defense and mucociliary protection.** While there is increased immune activity and inflammation, smoking also **impairs innate defense mechanisms**. These effects increase the risk, severity, and duration of infections. These effects include:



- Impaired ability of macrophages and natural killer cells to destroy bacteria and viruses.
- Impaired mucosal defenses, which can lead to bacterial colonization of the respiratory tract. Mucus secreted by goblet cells and submucosal muco-serous glands and containing trapped foreign particles is normally transported by the cilia toward the laryngopharynx (the mucociliary escalator) for either expectoration or swallowing.
- In COPD patients, smoking paralyzes the cilia and compromises this important defense mechanism. Importantly these cilia extend through much of the airways from the bronchi to bronchioles.

**Increased Neoplasia.** Cigarette smoke toxins are strongly implied in the development of many cancers (see list above). The mechanisms include mutations in tumor suppression genes (eg, p53) and oncogenes (eg, RAS). Benzopyrene, a constituent of cigarette smoke, forms metabolites that intercalate into DNA, interfering with normal gene transcription.

## How Do We Diagnose Patients Who May Develop Smoking–Related Lung Injury? (LO#4)

Given the long list of smoking consequences, diagnostic and treatment strategy should focus on **prevention and smoking cessation**. Studies have shown that only 42% of myocardial infarction smokers received smoking cessation interventions at hospital discharge, and only 21% of smokers received counseling at clinic visit. Thus we do not want to wait until disease symptoms develop. This need has led to emphasis on the history and on screening strategies.

## Smoking History

The first step is to obtain a good history from all of your patients, inquiring about current and past smoking habits at every clinic visit. Routinely asking about tobacco use has been shown to increase smoking cessation rates. Also ask about use of other tobacco and nicotine products such as vaping, cigars, pipes, chewing (smokeless) tobacco, and water pipes, since many tobacco users use a variety of tobacco products.

Ask about smoking frequency, products used, degree of dependence, when and how successfully they tried to quit, and which strategies they used (eg, use of nicotine gum).

Those who do not use tobacco products should be questioned about secondhand/passive smoke exposure (eg, in the home, car, or workplace). Such exposure can increase nonsmokers' health risks (eg, coronary heart disease and lung cancer).

## Screening Tests in Patients Who Smoke

Some patients who have smoked for a long time benefit from **secondary prevention** screening tests, which are performed even before symptoms arise. Such tests include:

- **Annual helical CT scan of the lungs.** This test is now recommended for those 50 to 80 years old who are either current smokers or have quit within the past 15 years and have at least a 20-pack-year smoking history. This has been found to lessen the risk of devastating lung cancer because it promotes early detection and treatment. Note that screening chest x-rays are not sensitive or specific enough to be used for routine prevention.

- **Screening spirometry (eg, FEV<sub>1</sub>/FVC).** This screening is performed on any smoking patient with respiratory symptoms and is recommended by some physicians for heavy smokers regardless of symptoms to detect early cases of COPD (prior to the onset of symptoms) and guide management.
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least 50-pack-years of smoking who are 20 to 80 years old.  
Lung CT scans are recommended for current/recent smokers with at

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## How Do We Help Patients Stop Smoking and Prevent Smoking-Related Illness? (LO#5)

The only truly effective method to prevent smoking-related illness is **abstinence from smoking**. Importantly, no amount of smoking is safe, so while smoking reduction is a step in the right direction, complete smoking cessation will yield the most health benefits. Quitting smoking will lower the elevated risk for smoking-associated illnesses. It also can improve the course

of existing smoking-related illnesses such as lung cancer and COPD. For example, smoking cessation improves cancer survival rates.

## Counseling

Initial counseling for tobacco cessation should follow the “5 A’s approach”: Ask, Advise, Assess, Assist, and Arrange, recommended by the US Agency for Healthcare Research and Quality. This strategy helps patients to stop smoking by using motivational interviewing principles.

1. **Ask**—Identify and document tobacco use status for every patient at every visit.
2. **Advise**—In a clear, strong, and personalized way, urge every tobacco user to quit.
3. **Assess**—Is the tobacco user willing to make a quit attempt at this time?
4. **Assist**—For the patient willing to make a quit attempt, use counseling and pharmacotherapy to help them quit.
5. **Arrange**—Follow up with the patient, in person or by telephone, ideally within the first week after the patient quits.

Patients also should be counseled about how their habit may affect their family and others via **second-hand smoke**. Exposure to second-hand smoke increases nonsmoker’s risks of lung cancer and ischemic heart disease. This is why most US regulations now prohibit indoor smoking in public and large private buildings (eg, apartments, theaters) as well as on all forms of public transportation. Some states also ban smoking in cars with minors present.

## Managing Nicotine Withdrawal

In patients who choose to quit smoking, the risk of recurrence is very high due to nicotine withdrawal symptoms such as decreased concentration,

insomnia, mood disturbances, and weight gain from increased appetite. So, to maintain smoking cessation, discuss treatment options that can attenuate withdrawal symptoms.

Nondrug strategies initially can include support groups and counseling, as well as advising the patient to occupy their hands and mouth with harmless cigarette substitutes such as sugarless gum, hard candy, straws, cinnamon sticks, or carrot sticks.

Some patients may benefit from pharmacologic therapy to reduce symptoms of withdrawal, there are multiple options available.

- Over-the-counter option: **Nicotine replacement therapy (NRT)**
  - The benefit of NRT is that it is readily accessible for patients, given it is an over-the-counter medication that is readily available. Also, it comes in a variety of dosage forms to best suit the patient's needs. This includes:
    - Patches
    - Gum
    - Lozenges
    - Nasal spray
    - Inhalers
  - Additionally, these dosage forms come in a variety of strengths. They are meant to be used and tapered so that the patient can ease the withdrawal symptoms and kick the habit.
- There are a couple of prescription/ non-nicotine options. These can be powerful aids in smoking cessation when they are carefully prescribed.
  - **Varenicline**
    - **Mechanism:** A partial agonist at nicotinic receptors. It both stimulates the receptor (reducing withdrawal symptoms and

cravings) and blocks nicotine from binding (reducing the rewarding effects of smoking).

- **Pros:** Highest rates of smoking cessation among monotherapies; reduces both cravings and the pleasurable effects of cigarettes.
- **Cons:** Requires dose titration; may be costly; some patients experience bothersome side effects.
- **ADRs:** Nausea (most common), insomnia, vivid dreams. Rare but serious concerns include mood changes, depression, or suicidal ideation—patients should be monitored for neuropsychiatric symptoms.

○ **Bupropion**

- **Mechanism:** A norepinephrine–dopamine reuptake inhibitor (NDRI) that also antagonizes nicotinic acetylcholine receptors. It reduces cravings and withdrawal symptoms.
- **Pros:** Can be helpful for patients with co-existing depression; does not cause weight gain and may even reduce post-cessation weight gain; available in generic form (less costly).
- **Cons:** Less effective than varenicline when used alone; requires adherence to daily dosing; contraindicated in patients with seizure disorders or eating disorders.
- **ADRs:** Insomnia, dry mouth, headache, agitation. The most serious risk is **seizure**, particularly in predisposed patients (e.g., those with seizure history, alcohol withdrawal).

 **CLINICAL CORRELATION**

When prescribing varenicline, it is important to set a specific quit date with the patient, usually within the first 1–2 weeks of therapy. Varenicline works best when started before quitting, as the partial agonist activity reduces cravings

during this lead-in period. Without a planned quit date, patients may continue smoking while on the medication, which increases nicotine exposure and can make side effects—particularly nausea—much more difficult to tolerate.

With non-nicotine therapies, patients may be completely weaned off nicotine over time. Note that while switching to vaping may be appealing to some (especially young) patients, it is not recommended as part of this strategy.

contain nicotine may be used, such as bupropion and varenicline. To help with nicotine withdrawal symptoms, agents that do not



## CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

Taking a look back at JB, how should you advise him?

You talk to JB about how tobacco smoke affects the body as well as the known risks of vaping. You also talk to him about how to stop smoking and offer him resources to aid in his journey to stop. For JB, who has smoked for several

years, the first step to quitting might be to use nicotine patches or gums, and the next step might be total weaning of nicotine. JB should continue to receive counseling to help him in his smoking cessation journey.

## Summary

- Smoking is an important modifiable risk factor for the development of many disorders, including cardiovascular, pulmonary, and neoplastic disease.
- Second-hand smoking is a public health issue arising from the high prevalence of smokers in society.
- Vaping is not safe and can cause nicotine toxicity and lung disease.
- Smokers have a three-fold heightened mortality rate compared to nonsmokers.
- Smoking causes chronic obstructive pulmonary disease (COPD) and lung cancer, and can worsen asthma.
- Smoking causes cardiovascular diseases such as myocardial infarction, stroke, and peripheral arterial disease.
- Smoking causes many types of cancer, including lung, gastrointestinal (GI), urinary, and laryngeal.
- Other smoking effects include acceleration of chronic kidney disease (CKD), eye disorders, ulcers, and increasing the risk of type 2 diabetes mellitus.
- Nicotine is the psychoactive substance contained in cigarettes and is the driver of smoking addiction.
- Nicotine has psychoactive effects on the brain's reward center, interfering with dopaminergic systems in the nucleus accumbens.



- Nicotine stimulates the sympathetic nervous system, leading to tachycardia, increased blood pressure, increased platelet aggregation, and loss of appetite.
- Tobacco smoke contains multiple chemicals that can increase vascular atherosclerosis and thrombosis, increase inflammation, impair the immune response, impair mucosal defenses, and cause cellular neoplasia and cancer.
- Diagnosis of smoking-related lung injury relies strongly on a thorough smoking history.
- Screening CT is recommended for smokers at highest risk.
- Primary treatment for all smoking-related illness is smoking cessation.
- Cessation is best accomplished by intensive counseling and education, supplemented by support groups, nicotine replacement therapies, non-nicotine medications, and harmless cigarette substitutes.
- The hazards of second-hand smoke should be discussed with all patients, but especially smokers and their families.

## Review Questions

1. Which of the following is present in tobacco smoke, but not in vaping smoke?

- A. Amphetamines
- B. Carbon monoxide
- C. Nicotine
- D. Propylene glycol

E. Vitamin E acetate

**Explanation** (requires correct answer)

2. Which of the following is a known health outcome of long-term nicotine usage?

- A. Nicotine poisoning
- B. Neoplasia formation
- C. Increased blood pressure
- D. Osteoporosis risk
- E. Alzheimer's prevention

**Explanation** (requires correct answer)

### 3. What is a common symptom of nicotine withdrawal?

- A. Increased appetite
- B. Dizziness
- C. Mood disturbance
- D. Rash
- E. Sneezing

**Explanation** (requires correct answer)

- A. Varenicline
- B. Bupropion
- C. Nicotine patches
- D. Nicotine gum
- E. Nicotine lozenges

**Explanation** (requires correct answer)

**5. Which test is recommended for heavy smokers to detect early COPD?**

- A. CT scan
- B. Screening Spirometry
- C. Chest x-ray
- D. Blood test
- E. MRI

Explanation (requires correct answer)

**How would you rate this Brick?**



## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Describe tobacco smoking and vaping and their epidemiology.
- 02 Identify tobacco-related adverse health outcomes.
- 03 Explain the pathophysiology of tobacco smoking-related disease.
- 04 Describe how to screen patients for tobacco use and exposure.
- 05 Outline management options for smoking cessation including pharmacological interventions and secondary prevention of smoking-related illness.

## Objective Confidence: 0/5